

Introduction to Managerial Accounting and Cost Concepts

1. Managerial vs. Financial Accounting

Two Branches of Accounting

Managerial Accounting:

- Provides information **for managers** of an organization
- Used by people who **direct and control** operations
- **Internal focus** - for decision makers inside the company

Financial Accounting:

- Provides information to **external parties**
- Users include: stockholders, creditors, investors, regulators
- **External focus** - for people outside the organization

Key Differences

Aspect	Financial Accounting	Managerial Accounting
1. Users	External persons making financial decisions	Internal managers who plan and control
2. Time Focus	Historical perspective (past events)	Future emphasis (planning ahead)
3. Verifiability vs Relevance	Emphasis on verifiability (must be proven)	Emphasis on relevance (useful for decisions)
4. Precision vs Timeliness	Emphasis on precision (exact figures)	Emphasis on timeliness (quick estimates OK)
5. Subject	Whole organization (company-wide)	Segments of organization (departments, products)
6. Requirements	MUST follow GAAP and prescribed formats	Need NOT follow GAAP or any prescribed format

GAAP = Generally Accepted Accounting Principles

Example Understanding the Differences

Financial Accounting Example:

- Annual report showing company made \$5,234,567 profit last year
- Must be verified by auditors

- Follows strict GAAP rules
- Shows entire company performance

Managerial Accounting Example:

- Report estimating Product A will generate \$500,000 next quarter
- Doesn't need external verification
- Can use any format management finds useful
- Shows specific product segment

2. Work of Management

Management performs three primary functions:

Function 1: Planning

Definition: Setting objectives and determining how to achieve them

Activities:

- Formulating long-term plans (strategic planning)
- Formulating short-term plans (operational planning)
- Setting goals and targets
- Budgeting resources

Example:

- Planning to increase sales by 15% next year

Function 2: Directing and Motivating

Definition: Mobilizing people to carry out plans and run routine operations

Activities:

- Assigning tasks
- Leading employees
- Motivating staff
- Day-to-day operations
- Implementing the plans

Example:

- Manager assigns sales territories

Function 3: Controlling

Definition: Ensuring that plans are being followed

Activities:

- Measuring performance
- Comparing actual results to planned results
- Taking corrective action when needed
- Identifying deviations from plan

Example:

- Comparing actual sales to budgeted sales

3. Planning and Control Cycle

The management process follows a continuous cycle:

1. Formulating Plans (PLANNING)



2. Implementing Plans (DIRECTING & MOTIVATING)



3. Measuring Performance (CONTROLLING)



4. Comparing Actual to Planned (CONTROLLING)



5. DECISION MAKING



Back to 1. (New Plans)

4. Comparing Merchandising and Manufacturing

Merchandising Companies (Retailers)

Activities:

- **Buy** finished goods from suppliers
- **Sell** finished goods to customers

- Do NOT manufacture anything

Example:

- MegaLoMart buys TVs from Samsung
- MegaLoMart sells TVs to customers

Manufacturing Companies

Examples: Toyota, Samsung, General Electric

Activities:

- **Buy** raw materials
- **Produce** (manufacture) finished goods
- **Sell** finished goods to customers

Example:

- Samsung buys raw materials (silicon, plastic, metals)
- Samsung manufactures TVs
- Samsung sells TVs to retailers or customers

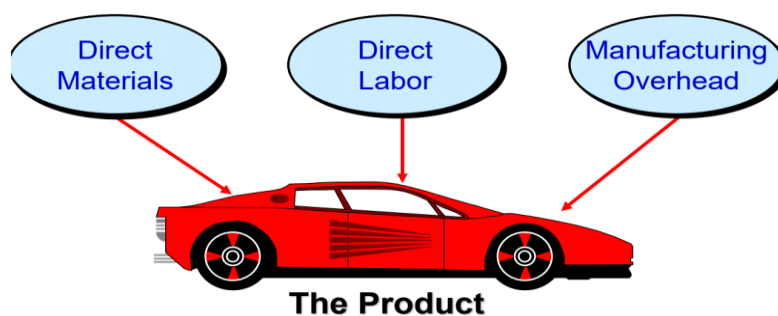
Key Difference

Merchandiser: Buy → Sell

Manufacturer: Buy Raw Materials → Produce → Sell

5. Manufacturing Costs

Manufacturing companies have three types of manufacturing costs:



Type 1: Direct Materials

Definition: Materials that become an integral part of the finished product and can be conveniently traced directly to it.

Characteristics:

- Physical part of the product
- Easy to trace to specific units
- Directly identifiable

Examples:

- **Automobile:** Steel, tires, radio, seats, glass
- **Furniture:** Wood, fabric, screws
- **Computer:** Circuit boards, hard drive, screen

Type 2: Direct Labor

Definition: Labor costs that can be **easily traced** to individual units of product.

Characteristics:

- Workers directly work on the product
- Time can be tracked to specific units
- "Touch labor" - physically work on product

Examples:

- **Automobile factory:** Assembly line workers installing parts
- **Bakery:** Bakers making bread

Non-Example (Indirect Labor):

- Factory supervisor (oversees but doesn't directly make product)
- Maintenance workers (support production but don't make product)

Type 3: Manufacturing Overhead

Definition: Manufacturing costs that **cannot be traced directly** to specific units produced.

Also called: Factory overhead, indirect manufacturing costs

Two Main Categories:**A. Indirect Materials**

- Materials used to support production
- Cannot be conveniently traced to specific units

- **Examples:**

- Lubricants for machines
- Cleaning supplies
- Small items like glue, tape, nails
- Factory supplies

B. Indirect Labor

- Wages paid to employees not directly involved in production

- **Examples:**

- Maintenance workers
- Security guards
- Quality inspectors
- Material handlers

Cost Classifications: Prime Cost and Conversion Cost

Prime Cost:

Prime Cost = Direct Materials + Direct Labor

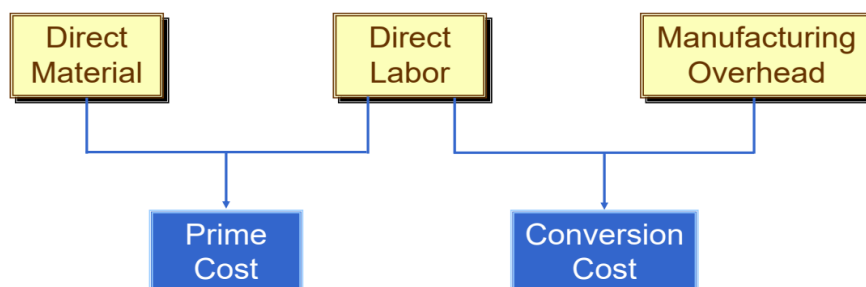
- The "prime" (main) costs of manufacturing
- Costs directly traceable to products

Conversion Cost:

Conversion Cost = Direct Labor + Manufacturing Overhead

- Costs to **convert** raw materials into finished products
- Labor and overhead needed for transformation

Visual Representation:



Example Calculation:

Given:

- Direct Materials: \$50,000
- Direct Labor: \$30,000
- Manufacturing Overhead: \$20,000

Prime Cost = \$50,000 + \$30,000 = **\$80,000**

Conversion Cost = \$30,000 + \$20,000 = **\$50,000**

Nonmanufacturing Costs

Marketing and selling costs . . .

❖ Costs necessary to get the order and deliver the product.

Administrative costs . . .

❖ All executive, organizational, and clerical costs.

6. Product Costs vs. Period Costs

Product Costs

Product costs are the costs directly related to manufacturing a product. These costs are attached to inventory and become expenses only when the product is sold.

What's Included:

- Direct Materials
- Direct Labor
- Manufacturing Overhead
- **ALL manufacturing costs**

Flow:

Product Costs → Inventory (Balance Sheet)



When Sold



Cost of Goods Sold (Income Statement)

Example:

- Manufacturing costs of \$100,000
- Initially recorded as inventory (asset)
- When products sold, becomes Cost of Goods Sold (expense)

Period Costs

Definition: Costs that are **NOT** included in product costs. They are **expensed immediately** on the income statement in the period incurred.

What's Included:**1. Marketing and Selling Costs**

- Advertising
- Sales commissions
- Delivery costs
- Costs to get orders and deliver products

2. Administrative Costs

- Executive salaries
- Accounting department
- Office supplies
- All organizational and clerical costs

Flow:

Period Costs → Immediately Expensed (Income Statement)

Example:

- Sales commissions of \$10,000
- Immediately recorded as expense in current period
- NOT held in inventory

Quick Check Example: Boeing

Question: Which costs are manufacturing overhead at Boeing?

A. Depreciation on factory forklift trucks ✓ B. Sales commissions X (Period cost - selling) C. Cost of flight recorder in Boeing 767 X (Direct material) D. Wages of production shift supervisor ✓

Answers: A and D

Explanations:

- **A (✓):** Factory forklift is manufacturing overhead (indirect)
- **B (X):** Sales commissions are selling costs (period costs)
- **C (X):** Flight recorder is direct material (part of plane, traceable)
- **D (✓):** Supervisor wages are indirect labor (overhead)

Quick Check Example: Period vs. Product Cost

Question: Which is a period cost rather than product cost?

A. Manufacturing equipment depreciation X (Product - overhead) B. Property taxes on corporate headquarters ✓ C. Direct materials costs X (Product - direct material) D. Electrical costs for production facility X (Product - overhead)

Answer: B

Explanation: Corporate headquarters is administrative, not manufacturing. This is a period cost.

7. Balance Sheet: Merchandiser vs. Manufacturer**Merchandiser Balance Sheet****Current Assets:**

- Cash
- Receivables
- Prepaid Expenses
- **Merchandise Inventory** (one inventory account)

Manufacturer Balance Sheet**Current Assets:**

- Cash
- Receivables
- Prepaid Expenses
- **Inventories** (three types):
 1. **Raw Materials**
 2. **Work in Process**
 3. **Finished Goods**

Three Types of Manufacturing Inventory

1. Raw Materials Inventory

- **Definition:** Materials waiting to be processed
- Not yet put into production
- Still in storage/warehouse

Example:

- Steel plates waiting to be used
- Wood planks in lumber yard
- Fabric rolls waiting to be cut

2. Work in Process Inventory (WIP)

- **Definition:** Partially completed products
- Some material, labor, or overhead has been added
- Not yet finished

Example:

- Car with engine installed but no doors
- Half-sewn dress
- Partially assembled computer

3. Finished Goods Inventory

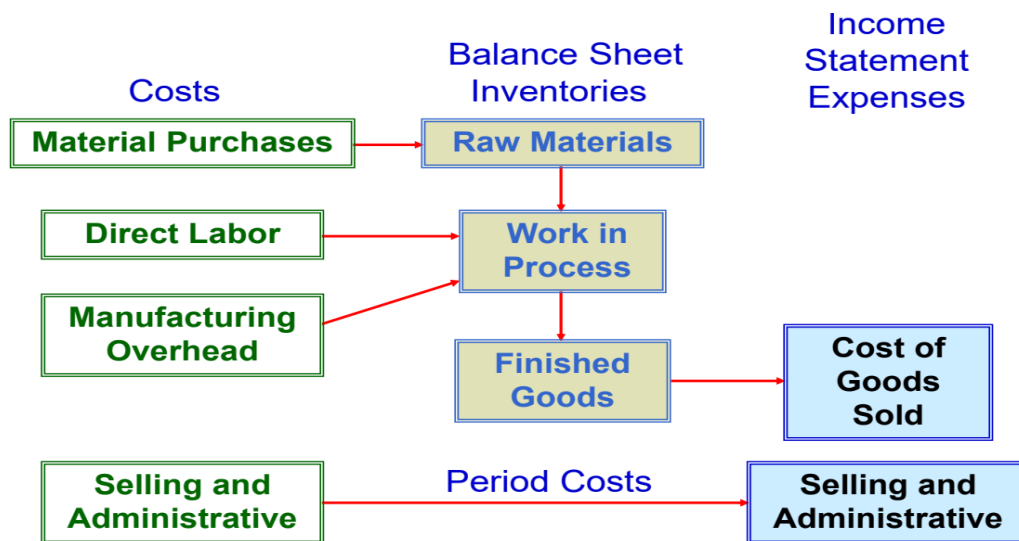
- **Definition:** Completed products awaiting sale
- Fully manufactured
- Ready for customers

Example:

- Completed cars in dealer lot
- Finished dresses on rack
- Boxed computers in warehouse

8. Cost Flows in Manufacturing

The Flow Diagram



Income Statement Comparison

Merchandising Company:

Cost of Goods Sold:

Beginning Merchandise Inventory	\$ 14,200
+ Purchases	234,150
= Goods Available for Sale	\$248,350
- Ending Merchandise Inventory	(12,100)
= Cost of Goods Sold	\$236,250

Manufacturing Company:

Cost of Goods Sold:

Beginning Finished Goods Inventory	\$ 14,200
+ Cost of Goods Manufactured	234,150
= Goods Available for Sale	\$248,350
- Ending Finished Goods Inventory	(12,100)
= Cost of Goods Sold	\$236,250

Key Difference: Manufacturers use "Cost of Goods Manufactured" instead of "Purchases"

9. Calculating Manufacturing Costs - Step by Step

The Inventory Flow Formula

General Formula:

Beginning Balance

+ Additions

= Available

- Ending Balance

= Withdrawals

This formula applies to ALL inventory accounts!

Step 1: Raw Materials Used

Beginning Raw Materials Inventory

+ Raw Materials Purchased

= Raw Materials Available for Use

- Ending Raw Materials Inventory

= Raw Materials Used in Production (Direct Materials)

Example:

Beginning Raw Materials: \$32,000

Purchases: \$276,000

Ending Raw Materials: \$28,000

Calculation:

$\$32,000 + \$276,000 = \$308,000$ available

$\$308,000 - \$28,000 = \$280,000$ used

Step 2: Total Manufacturing Costs

Direct Materials (from Step 1)

+ Direct Labor

+ Manufacturing Overhead

= Total Manufacturing Costs

Example:

Direct Materials Used: \$280,000

Direct Labor: \$375,000

Manufacturing Overhead: \$180,000

Total Manufacturing Costs = \$280,000 + \$375,000 + \$180,000

= \$835,000

Step 3: Cost of Goods Manufactured

Beginning Work in Process Inventory

+ Total Manufacturing Costs (from Step 2)

= Total Work in Process

- Ending Work in Process Inventory

= Cost of Goods Manufactured

Example:

Beginning WIP: \$125,000

Total Manufacturing Costs: \$835,000

Ending WIP: \$200,000

Calculation:

$\$125,000 + \$835,000 = \$960,000$ total WIP

$\$960,000 - \$200,000 = \$760,000$ cost of goods manufactured

Step 4: Cost of Goods Sold

Beginning Finished Goods Inventory

+ Cost of Goods Manufactured (from Step 3)

= Cost of Goods Available for Sale

- Ending Finished Goods Inventory

= Cost of Goods Sold

Example:

Beginning Finished Goods: \$130,000

Cost of Goods Manufactured: \$760,000

Ending Finished Goods: \$150,000

Calculation:

$\$130,000 + \$760,000 = \$890,000$ available

$\$890,000 - \$150,000 = \$740,000$ cost of goods sold

Complete Example Flow

Given Data:

- Beginning Raw Materials: \$32,000
- Purchases: \$276,000
- Ending Raw Materials: \$28,000
- Direct Labor: \$375,000
- Manufacturing Overhead: \$180,000
- Beginning WIP: \$125,000
- Ending WIP: \$200,000
- Beginning Finished Goods: \$130,000
- Ending Finished Goods: \$150,000

Step-by-Step Solution:

Step 1: Direct Materials Used

$\$32,000 + \$276,000 - \$28,000 = \$280,000$

Step 2: Total Manufacturing Costs

$\$280,000 + \$375,000 + \$180,000 = \$835,000$

Step 3: Cost of Goods Manufactured

$\$125,000 + \$835,000 - \$200,000 = \$760,000$

Step 4: Cost of Goods Sold

$$\$130,000 + \$760,000 - \$150,000 = \$740,000$$

10. Cost Behavior: Variable and Fixed Costs

Understanding Cost Behavior

Cost Behavior: How a cost will react to changes in the level of business activity.

Why It Matters:

- Planning and budgeting
- Decision making
- Cost prediction
- Profitability analysis

Variable Costs

Definition: Costs that change in total as activity level changes, but remain constant per unit.

Total Variable Cost Behavior:

As Activity Increases → Total Variable Cost Increases

As Activity Decreases → Total Variable Cost Decreases

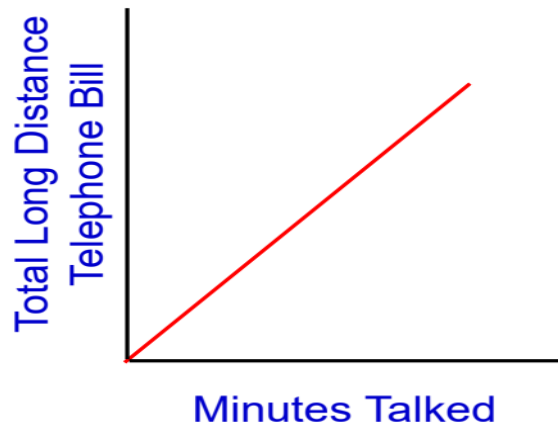
Variable Cost Per Unit:

Remains CONSTANT regardless of activity level

Example: Long Distance Telephone Bill

Suppose: \$0.10 per minute

Minutes Talked	Cost Per Minute	Total Cost
100	\$0.10	\$10.00
200	\$0.10	\$20.00
500	\$0.10	\$50.00



Fixed Costs

Definition: Costs that remain **constant in total** regardless of activity level, but **per unit cost decreases** as activity increases.

Total Fixed Cost Behavior:

Remains CONSTANT regardless of activity level

Fixed Cost Per Unit:

As Activity Increases → Cost Per Unit Decreases

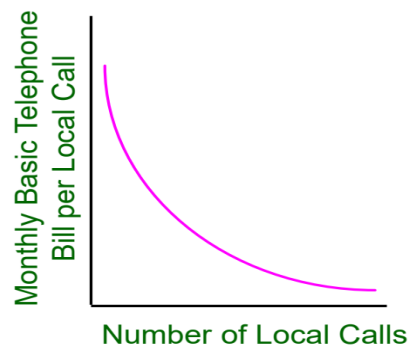
As Activity Decreases → Cost Per Unit Increases

Example: Monthly Basic Telephone Bill

Suppose: \$30 per month (allows unlimited local calls)

Local Calls	Total Monthly Bill	Cost Per Call
100	\$30	\$0.30
200	\$30	\$0.15
600	\$30	\$0.05

Graph:



Behavior of Cost (within the relevant range)		
Cost	In Total	Per Unit
Variable	Total variable cost changes as activity level changes.	Variable cost per unit remains the same over wide ranges of activity.
Fixed	Total fixed cost remains the same even when the activity level changes.	Fixed cost per unit goes down as activity level goes up.

11. Cost Classifications for Decision Making

Direct vs. Indirect Costs

Direct Costs:

- Can be **easily and conveniently traced** to a specific cost object
- Cost object = anything we want to measure cost of (product, department, project)

Examples:

- Direct materials in a product
- Direct labor on a specific job
- Salary of department manager (direct to that department)

Indirect Costs:

- **Cannot be easily traced** to a specific cost object
- Must be allocated using some method

Examples:

- Manufacturing overhead
- Factory rent (shared by all products)
- CEO salary (benefits entire company)

Differential Costs and Revenues

Definition: Costs and revenues that **differ among alternatives**.

Also called:

- Relevant costs
- Incremental costs

- Avoidable costs

Example:

You have two job offers:

Job	Salary (Revenue)	Commuting Cost
Hometown Job	\$1,500/month	\$0
City Job	\$2,000/month	\$300/month

Differential Revenue:

$$\$2,000 - \$1,500 = \$500$$

Differential Cost:

$$\$300 - \$0 = \$300$$

Net Differential Benefit:

$$\$500 - \$300 = \$200 \text{ (City job is better by \$200/month)}$$

Relevant vs. Irrelevant Costs

Relevant costs are future costs that differ between alternatives and therefore affect decision-making.

Relevant Costs:

- Future costs
- Differ between alternatives
- Affect the decision

Irrelevant Costs:

Irrelevant costs are costs that do not affect decisions because they remain the same regardless of the alternative chosen

- Past costs (sunk costs)
- Same for all alternatives
- Don't affect the decision

Decision Examples: Drive or Take Train to Portland?

Scenario: Deciding whether to drive or take train to a concert

Question 1: Is the pizza you ate last night relevant?

Answer: NO X

Reason: Sunk cost - already spent, doesn't differ between alternatives

Question 2: Is the train ticket cost relevant?

Answer: YES ✓

Reason:

- Future cost
- Only incurred if you take the train
- Differs between alternatives

Question 3: Is the annual car license fee relevant?

Answer: NO X

Reason:

- Already paid (or must pay regardless)
- Same whether you drive or not
- Doesn't differ between alternatives

Question 4: Is car depreciation relevant?

Answer: NO X

Reason:

- Car depreciates whether you drive or not
- Not a differential cost
- Historical/sunk cost

Question 5: What if you could sell the car for \$5,000?

Answer: \$5,000 is NOT a sunk cost - it's relevant!

Reason:

- Selling price is a future benefit
- It's an opportunity cost if you keep the car

- Relevant to decision

Opportunity Costs

Definition: The potential benefit that is **given up** when one alternative is selected over another.

Characteristics:

- NOT recorded in accounting books
- Very important for decision making
- Represents foregone benefits

Example 1: College Decision

If you attend college:

- You give up working full-time
- Could earn \$15,000 per year working

Opportunity Cost of College: \$15,000 per year (foregone earnings)

Example 2: Using a Building

You own a building you could rent for \$20,000/year:

- Use it for your own business: Opportunity cost = \$20,000
- Rent it out: Opportunity cost = business profits you gave up

Important: Opportunity costs are **real economic costs** even though not recorded in books!

Sunk Costs

Definition: Costs that have already been incurred and **cannot be changed** by any decision.

Characteristics:

- Past costs
- Cannot be undone
- Should be IGNORED in decisions
- NOT differential costs

Example 1: Car Purchase

Two years ago you bought a car for \$10,000:

- The \$10,000 is SUNK

- You can't change it whether you:
 - Drive the car
 - Park it
 - Trade it
 - Sell it

The \$10,000 is irrelevant to future decisions!

Example 2: Movie Ticket

You bought a \$15 movie ticket but the movie is terrible:

- The \$15 is SUNK
- Should you stay or leave?
- The \$15 shouldn't affect your decision
- Only consider: enjoyment of staying vs. what else you could do

Common Mistake: People often consider sunk costs ("I already paid for it!") when they shouldn't.

Example 3: Is selling price a sunk cost?

If your car could be sold now for \$5,000:

- Answer: NO, not a sunk cost
- \$5,000 is a potential future benefit
- Very relevant to decisions!